

MODULE 3 – SELENIUM PYTHON & BEHAVE

Assignment Report

Student	Ishani Mandal
Program	B.Tech – Computer Science and Information Technology
Module	Module 3
Technology	Python, Selenium WebDriver, Behave BDD
Browser	Google Chrome
Python	3.14.x

1. Overview

This report documents the Module 3 Selenium automation work completed using Python, Selenium WebDriver and the Behave BDD framework. The assignments cover web automation, multiple-element handling, Page Object Model, Gherkin scenarios and data-driven testing.

2. Assignment 1 – Selenium Python and Behave BDD

Scenario: Implementation of Selenium Python and Behave BDD.

Objective: Open SauceDemo, enter valid credentials, click Login and verify successful login.

Test data: standard_user / secret_sauce.

Verification: The current URL was checked for inventory.html.

Result: 1 feature passed, 1 scenario passed and 4 steps passed.

3. Assignment 2 – Multiple Element Identification

Objective: Identify multiple elements of the same type and work with the resulting list.

Implementation: Selenium find_elements() with By.TAG_NAME and the a tag was used to collect links on the SauceDemo login page. The total count and link text were printed.

Observed result: Total links: 0. The browser opened successfully and remained open until Enter was pressed.

Key concept: find_elements() returns a list of matching WebElement objects and an empty list when no elements match.

4. Assignment 3 – Selenium Page Object Model with Behave

Scenario: Selenium Page Object Model in the Python Behave Framework.

Objective: Use Behave and Gherkin for a test-data-driven approach while separating page locators and actions through the Page Object Model.

Project structure: pages/login_page.py, features/login.feature, features/steps/login_steps.py, features/environment.py and requirements.txt.

Page Object: LoginPage contains username, password and login-button locators and methods for entering credentials and clicking Login.

Data-driven scenarios: standard_user/secret_sauce, locked_out_user/secret_sauce and invalid_user/wrong_pass.

Expected validations: successful login reaches inventory.html; locked and invalid users display the corresponding application error.

5. Environment and Dependencies

Python 3.14.x

Selenium 4.49.0

Behave 1.3.3

Google Chrome

6. Learning Outcomes

- Created Selenium WebDriver automation using Python.
- Implemented BDD scenarios using Behave and Gherkin.
- Used find_elements() for multiple-element identification.
- Applied Page Object Model to separate page actions from test steps.
- Implemented data-driven testing using a Behave Scenario Outline.
- Verified different login outcomes using assertions and application messages.
- Used isolated virtual environments and requirements.txt files.

7. Assignment Status

Assignment 1 – Completed

Assignment 2 – Completed

Assignment 3 – Implementation completed; final execution verification is pending in the current record.

Assignment 4 – Not yet documented.

Note: This report is based on the Module 3 assignment details and execution information available in the project record. Assignment 4 is not presented as completed because its actual task and execution result have not yet been provided.